

Optimal Construction of Subband Coders Using Lloyd–Max Quantizers

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Abstract—A new method is presented for the analysis of the effects of Lloyd–Max quantization in subband filterbanks and for the optimal design of such filterbanks. A rigorous statistical model of a vector Lloyd–Max quantizer is established first, consisting of a linear time-invariant filter followed by additive noise uncorrelated with the input. On the basis of this model, an expression for this variance of the error of a subband coder using Lloyd–Max quantizers is explicitly determined. Given analysis filters that statistically separate the subbands, it is shown that this variance is minimized if the synthesis filters are chosen, which would achieve perfect reconstruction in lossless coding. The globally optimum such filterbank, minimizing the coder error variance, is further obtained by proper choice of its analysis filters. An alternative design method is also evaluated and optimized. In this, the errors correlated with the signal are set to zero, leaving a random error residue uncorrelated with the signal. This design method is optimized by choosing the analysis filters so as to minimize the random error variance. The results are evaluated experimentally in the realistic setting of a logarithmically split subband image coding scheme.

Index Terms—Filterbanks, Lloyd–Max quantization, subband coding.

I. INTRODUCTION

ANALYSIS/SYNTHESIS filterbanks have been widely used to split up signals into frequency bands for the purpose of coding each subband using a coder and bit rate matched to the statistics of this particular band. If the analysis/synthesis filters are properly designed, and if the signal is coded losslessly, the subbands may then be reassembled at the receiver, producing a perfect replica of the original signal. However, in many applications lossless coding produces insufficient compression, necessitating the use of lossy schemes, which involve signal quantization for the coding of each subband. The resulting errors, due to quantization, will produce nonnegligible aliasing and distortion error terms in the reconstructed signal. As a result, the need exists to analyze the subband and quantization system as a whole, with a view toward the minimization of aliasing and distortion due to quantization noise, by appropriate choice of the analysis and synthesis filters.

The traditional approach in the literature has been to address the problems of filterbank selection and quantizer or

transmission noise elimination, separately. The purpose of the classical methodology in [1] as well as much of the subsequent work [2] is to optimize the bit allocation in subbands with preselected analysis and synthesis filters. Recently, there has been increased interest in developing methodologies for the solution of the composite problem of optimizing the subband decomposition of quantized signals, by appropriately selecting the analysis and synthesis structures in relation to the quantizers used. In [3], a method is presented for the optimal signal reconstruction from its quantized version in nonorthogonal subband coding schemes. In [4] the coding gain is maximized by proper selection of the wavelet scaling functions for perfect reconstruction filterbanks based on biorthogonal wavelet decomposition. In [5] statistical models are used for the design of analysis/synthesis systems for quantized signals assuming overlap-add synthesis.

Lloyd–Max quantizers are specifically examined in [6]. On the basis of both analytical [1] and experimental results, the Lloyd–Max quantizer is modeled by a nonunity constant gain, plus additive random noise. Subsequently, the quantization errors in subband coding using quadrature mirror filters (QMF's) are investigated, and the consequent aliasing and distortion errors in the reconstructed signal are evaluated using deterministic analysis and extensive experimental verification. Cyclostationary modeling is used to derive the optimal compensation of such quantization errors in [7].

In [8], the same model of Lloyd–Max quantizers is adopted for the deterministic analysis of general subband coding schemes. It is further argued that an appropriately scaled version of the synthesis filters of a perfect reconstruction filterbank will produce a reconstruction error that is uncorrelated with the input signal and therefore can be dealt with using random noise removal methods.

A first purpose of the present paper is the formulation of a more general statistical description of the errors in subband coding due to Lloyd–Max signal quantization. First, a general model is constructed of a vector Lloyd–Max quantizer, consisting of a linear time-invariant filter plus additive noise, which is uncorrelated with the quantizer input sequence. This generalizes the classical gain-plus additive noise description [1] of the Lloyd–Max quantizer, which was exploited in [6] and [8]. The methodology developed in [10] is then used to obtain an explicit expression for the variance of the overall error difference, i.e., the difference between the original and the reconstructed signal when Lloyd–Max quantizers are used in all subbands of the filterbank.

Subsequently, two strategies are developed for the optimization of the filterbank design. In the first, a minimization of the

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overall error variance is sought. It is shown that given arbitrary analysis filters, the synthesis filters achieving this minimization are those composing a perfect reconstruction filterbank with the analysis filters. The analysis filters of the filterbank are then chosen so as to achieve minimum overall error variance. In the second strategy, the approach in [6] and [8] is followed. In this, all errors correlated with the input are set to zero, leaving a random error residue. This approach is optimized by choosing the analysis and synthesis filters so as to minimize the variance of the resulting random error. In both approaches the optimum bit allocations to the subbands are calculated using the methodology in [11].

The methods developed are experimentally evaluated in application to image coding. The performance of the above two strategies is determined, first using the optimal for each strategy filterbanks, and then with the use of high-performance filterbanks proposed in [8], [20] and [28], and evaluated in [27]. It is confirmed that the optimally constructed filterbanks achieve with the same bit rate superior encoded image quality even when used with simple scalar Lloyd–Max quantizers.

Methods for the statistical analysis of filterbanks using vector Lloyd–Max quantizers are established in Section II. The first above strategy of minimizing the overall reconstruction error is optimized in Section III, while the second strategy is optimized in Section IV. The results are generalized for N -dimensional signals with arbitrary sampling matrices and M -channel filter banks in Section V. The experimental results are presented in Section VI and conclusions and recommendations for further work are included in Section VII.

II. STATISTICAL ANALYSIS OF FILTERBANKS USING VECTOR LLOYD–MAX QUANTIZERS

Consider the subband filterbank in Fig. 1 and assume momentarily that the decimator output is not quantized or otherwise corrupted by external noise. In compact notation, for each subband k , $k = 0, 1$, the output of the prefilter followed by decimation and interpolation of zeros will then be given (Fig. 1) by

$$v_k[m] = c[m] \sum_i h_k[i] x[m - i] = c[m] u_k[m] \quad (1)$$

where

$$c[m] = \frac{(1 + (-1)^m)}{2}. \quad (2)$$

Following the development in [10], we note that the sequence $v_k[m]$ is not wide-sense stationary; however, the time averages

$$\begin{aligned} R_{v_k x}[p] &= \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K v_k[m] x[m-p], \\ R_{v_k}[p] &= \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K v_k[m] v_k[m-p] \end{aligned} \quad (3)$$

are seen to exist (Appendix), under nonrestrictive conditions on $x[m]$. With the above definition, (3) gives in terms of the

autocorrelation function $R_x[p]$ of the input

$$\begin{aligned} R_{v_k x}[p] &= \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K c[m] u_k[m] x[m-p] \\ &= \frac{1}{2} \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K u_k[m] x[m-p] \\ &\quad + \frac{1}{2} \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K (-1)^m u_k[m] x[m-p]. \end{aligned}$$

By using Lemma 1 in the Appendix, the second above term is trivially seen to be zero. Therefore

$$R_{v_k x}[p] = \frac{1}{2} R_{u_k x}[p] = \frac{1}{2} \sum_i h_k[i] R_x[m-i] = \frac{1}{2} h_k * R_x$$

where the asterisk $*$ denotes convolution. Also, using the same Lemma

$$\begin{aligned} R_{v_k}[p] &= \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K c[m] c[m-p] u_k[m] u_k[m-p] \\ &= c[p] \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{m=-K}^K c[m] u_k[m] u_k[m-p] \\ &= \frac{1}{2} c[p] R_{u_k}[p] \end{aligned}$$

because $c[m]c[m-p] = c[p]c[m]$. Thus,

$$R_{v_k}[p] = \frac{1}{2} c[p] (h_k[p] * R_x[p] * h_k[-p]) \quad (4)$$

where $c[p]$ is given by (2). The corresponding Z -transforms are therefore related by

$$\Phi_{v_k x}(z) = \frac{1}{2} H_k(z) \Phi_x(z) \quad (5)$$

$$\begin{aligned} \Phi_{v_k}(z) &= \frac{1}{4} (\Phi_{u_k}(z) + \Phi_{u_k}(-z)) = \frac{1}{4} (H_k(z) \Phi_x(z) H_k(z^{-1}) \\ &\quad + H_k(-z) \Phi_x(-z) H_k(-z^{-1})). \end{aligned} \quad (6)$$

Similarly,

$$\begin{aligned} \Phi_{v_0 v_1}(z) &= \frac{1}{4} (H_0(z) \Phi_x(z) H_1(z^{-1}) \\ &\quad + H_0(-z) \Phi_x(-z) H_1(-z^{-1})). \end{aligned} \quad (7)$$

Let us now assume that a vector Lloyd–Max quantizer is used for each subband. An input-output model of such a quantizer supported by theoretical considerations and also by experimental data such as contained in [6] and [8] and also in Section VI of the present paper, is

$$s[m] = a[m] * v[m] + r[m] \quad (8)$$

where v, s are, respectively, the input and output of the quantizer, the asterisk denotes convolution, $a[m] = a[-m]$ is a

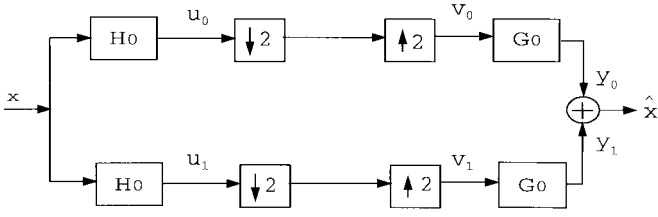


Fig. 1. Lossless encoder.

real sequence and $r[m]$ is an additive noise term uncorrelated with the input $v[m]$.

This model, developed in Lemma 2 in the Appendix is a generalization of the model in [6] and [8] constructed to produce noise sequences $\{r[m]\}$ truly uncorrelated with signal sequences $\{v[m]\}$, rather than simply uncorrelated terms $r[m], v[m]$ for the same m as does the model in [6] and [8]. As will be seen in the sequel, this model allows us to work in the frequency domain and thus to produce explicit expressions for the error variances in terms of the analysis and synthesis filters. The most suitable analysis and synthesis filters may then be easily selected using these expressions.

Since the quantizers are located after the subsamplers, (see Fig. 2), relation (8) will involve the subsampled signals

$$\tilde{v}_k[m] = u_k[2m], \quad k = 0, 1$$

and will have as outputs

$$\tilde{s}_k[m] = a[m] * \tilde{v}_k[m] + \tilde{r}_k[m].$$

From Lemma 2 in the Appendix

$$A_k(z) = A_k(z^{-1}) = \Phi_{\tilde{s}_k}(z)/\Phi_{\tilde{v}_k}(z) = 1 - \Phi_{\tilde{q}_k}(z)/\Phi_{\tilde{v}_k}(z) \quad (9)$$

where $\Phi_{\tilde{q}_k}(z)$ is the power spectrum of the quantizer error $\tilde{q}_k[m] = \tilde{v}_k[m] - \tilde{s}_k[m]$. Also, the noise spectrum is given by

$$\Phi_{\tilde{r}_k}(z) = A_k(z)(1 - A_k(z))\Phi_{\tilde{v}_k}(z). \quad (10)$$

The quantizers are physically located before the upsamplers as shown in Fig. 2. For the sake of ease of analysis and of notational simplicity, however, we shall equivalently place them after the upsamplers, assuming simply that the zeros interpolated by the upsamplers are always quantized to zero (Fig. 3). Now it is easily seen that the autocorrelations of $s_k[m], v_k[m], r_k[m]$ defined as in (3) are given by

$$\begin{aligned} R_{s_k}[p] &= R_{\tilde{s}_k}[p/2]c[p], & R_{v_k}[p] &= R_{\tilde{v}_k}[p/2]c[p], \\ R_{r_k}[p] &= R_{\tilde{r}_k}[p/2]c[p] \end{aligned}$$

where $c[p]$ is given by (2). Hence

$$\Phi_{s_k}(z) = \Phi_{\tilde{s}_k}(z^2), \quad \Phi_{v_k}(z) = \Phi_{\tilde{v}_k}(z^2), \quad \Phi_{r_k}(z) = \Phi_{\tilde{r}_k}(z^2)$$

and therefore, (9) and (10) become

$$\Phi_{s_k}(z) = A_k(z^2)\Phi_{v_k}(z) \quad (11)$$

$$\Phi_{r_k}(z) = A_k(z^2)(1 - A_k(z^2))\Phi_{v_k}(z). \quad (12)$$

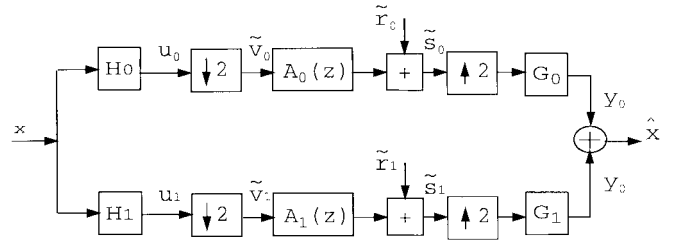


Fig. 2. Encoder with Lloyd-Max quantizers.

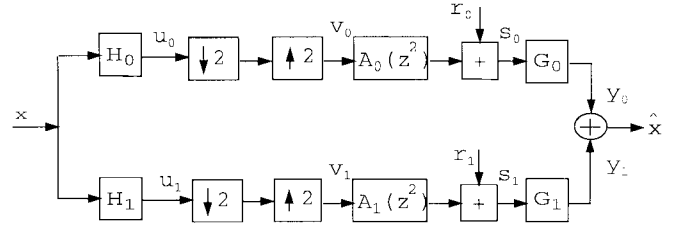


Fig. 3. Equivalent description of the encoder with Lloyd-Max quantizers.

Finally from Fig. 3, for $k = 0, 1$, using the orthogonality of $r_k[m]$ and $v_k[m]$ and (5)

$$\begin{aligned} \Phi_{s_k x}(z) &= A_k(z^2)\Phi_{v_k x}(z) + \Phi_{r_k x}(z) \\ &= \frac{1}{2}A_k(z^2)H_k(z)\Phi_x(z) + \Phi_{r_k x}(z) \\ \Phi_{s_k}(z) &= A_k^2(z^2)\Phi_{v_k}(z) + \Phi_{r_k}(z). \end{aligned}$$

Hence

$$\begin{aligned} \Phi_{y_k x}(z) &= G_k(z)\Phi_{s_k x}(z) \\ &= \frac{1}{2}G_k(z)A_k(z)H_k(z)\Phi_x(z) + G_k(z)\Phi_{r_k x}(z) \quad (13) \end{aligned}$$

$$\begin{aligned} \Phi_{y_k}(z) &= G_k(z)\Phi_{s_k}(z)G_k(z^{-1}) \\ &= G_k(z)(A_k^2(z^2)\Phi_{v_k}(z) + \Phi_{r_k}(z))G_k(z^{-1}) \quad (14) \end{aligned}$$

$$\begin{aligned} \Phi_{y_k y_l}(z) &= G_k(z)(A_k(z^2)A_l(z^2)\Phi_{v_k v_l}(z) \\ &\quad + \Phi_{r_k r_l}(z))G_l(z^{-1}), \quad k, l = 0, 1 \quad (15) \end{aligned}$$

where $\Phi_{v_k}(z)$ are given by (6), $\Phi_{r_k}(z)$ are given by (12), $\Phi_{r_0 r_1}(z)$ is the cross power spectrum of the noise terms, and $\Phi_{v_0 v_1}(z)$ is given by (7).

The sum of the final outputs y_0, y_1 should approximate optimally the input x . The reconstruction error variance is

$$\begin{aligned} E &= E\{(y_0[m] + y_1[m] - x[m])^2\} \\ &= R_{y_0}[0] + R_{y_1}[0] + R_x[0] + 2R_{y_0 y_1}[0] \\ &\quad - 2R_{y_0 x}[0] - 2R_{y_1 x}[0]. \end{aligned} \quad (16)$$

Hence

$$\begin{aligned} 2\pi j E &= \oint z^{-1} dz (\Phi_{y_0}(z) - \Phi_{y_0 x}(z) - \Phi_{x y_0}(z) + \Phi_{y_1}(z) \\ &\quad - \Phi_{y_1 x}(z) - \Phi_{x y_1}(z) + \Phi_{y_0 y_1}(z) \\ &\quad + \Phi_{y_1 y_0}(z) + \Phi_x(z)) \end{aligned}$$

and using (13)–(15)

$$\begin{aligned}
& 2\pi jE \\
&= \oint G_0(z)G_0(z^{-1})[A_0^2(z^2)\Phi_{v_0}(z) + \Phi_{r_0}(z)]z^{-1}dz \\
&+ \oint G_1(z)G_1(z^{-1})[A_1^2(z^2)\Phi_{v_1}(z) + \Phi_{r_1}(z)]z^{-1}dz \\
&+ \oint G_0(z)G_1(z^{-1})[A_0(z^2)A_1(z^2)\Phi_{v_0v_1}(z) \\
&+ \Phi_{r_0r_1}(z)]z^{-1}dz + \oint G_0(z^{-1})G_1(z) \\
&\times [A_0(z^2)A_1(z^2)\Phi_{v_0v_1}(z^{-1}) + \Phi_{r_0r_1}(z^{-1})]z^{-1}dz \\
&- \oint G_0(z)A_0(z^2)H_0(z)\Phi_x(z)z^{-1}dz \\
&- \oint G_1(z)A_1(z^2)H_1(z)\Phi_x(z)z^{-1}dz \\
&- 2 \oint G_0(z)\Phi_{r_0x}(z)z^{-1}dz - 2 \oint G_1(z) \\
&\times \Phi_{r_1x}(z)z^{-1}dz + \oint \Phi_x(z)z^{-1}dz. \quad (17)
\end{aligned}$$

Therefore

$$2\pi jE = 2\pi jE_R + 2\pi jE_F \quad (18)$$

where

$$\begin{aligned}
2\pi jE_R &= \oint G_0(z)G_0(z^{-1})\Phi_{r_0}(z)z^{-1}dz \\
&+ \oint G_1(z)G_1(z^{-1})\Phi_{r_1}(z)z^{-1}dz \\
&+ 2 \oint G_0(z)G_1(z^{-1})\Phi_{r_0r_1}(z)z^{-1}dz \\
&- 2 \oint G_0(z)\Phi_{r_0x}(z)z^{-1}dz \\
&- 2 \oint G_1(z)\Phi_{r_1x}(z)z^{-1}dz \quad (19)
\end{aligned}$$

is the error due to the random noise, while the error correlated with the input is composed of the remaining terms in (17).

We summarize the above development in the form of a theorem.

Theorem 1: The overall variance E of the reconstruction error $\hat{x}[m] - x[m]$ of the filterbank in Fig. 3 with analysis and synthesis filters $H_k, G_k, k = 0, 1$, respectively and with Lloyd–Max quantizers characterized by filter transfer functions $A_0(z^2), A_1(z^2)$ is given by (17). The variance E_R of the part of the reconstruction error which is uncorrelated with the input is given by (19). ■

Two strategies may now be formulated for the selection of the optimum analysis and synthesis filters. The first is to design the filter bank so as to minimize the overall reconstruction error (17). A second is to choose the suboptimal methodology proposed in [8] so as to set to zero the errors that are correlated with the input, leaving the minimization of the remaining error E_R for a later stage. We shall explore the first strategy next.

III. STRATEGY 1: MINIMIZATION OF THE OVERALL RECONSTRUCTION ERROR

We shall first assume that the analysis filters H_0, H_1 are chosen so as to statistically decouple the two subbands

$$\begin{aligned}
\Phi_{v_0v_1}(z) &= H_0(z)H_1(z^{-1})\Phi_x(z) \\
&+ H_0(-z)H_1(-z^{-1})\Phi_x(-z) = 0. \quad (20)
\end{aligned}$$

As a result of this, the cross-power spectrum of the noise terms may also be postulated to be zero, as follows:

$$\Phi_{r_0r_1}(z) = 0. \quad (21)$$

For the same reason, $\Phi_{r_0v_1}(z) = 0, \Phi_{r_1v_0}(z) = 0$ and hence

$$\Phi_{r_kx}(z) = 0, \quad k = 0, 1. \quad (22)$$

For a formal proof of the above, see Lemma 3 in the Appendix. Replacing the above, and also (12) in (17) gives

$$\begin{aligned}
2\pi jE &= \oint \Phi_x(z)z^{-1}dz \\
&+ \sum_{k=0}^1 \oint \left[\frac{1}{2}G_k(z)G_k(z^{-1})P_k(z)A_k(z^2) \right. \\
&\left. - G_k(z)H_k(z)\Phi_x(z)A_k(z^2) \right] z^{-1}dz \quad (23)
\end{aligned}$$

where

$$\begin{aligned}
P_k(z) &= 2\Phi_{v_k}(z) = \frac{1}{2}[H_k(z)\Phi_x(z)H_k(z^{-1}) \\
&+ H_k(-z)\Phi_x(z)H_k(-z^{-1})]. \quad (24)
\end{aligned}$$

On replacing (24) into (23), regrouping terms, and changing integration variables, we easily obtain the following equivalent expression

$$\begin{aligned}
2\pi jE &= \oint \Phi_x(z)z^{-1}dz \\
&+ \sum_{k=0}^1 \oint \left[\frac{1}{2}H_k(z)H_k(z^{-1})Q_k(z)\Phi_x(z)A_k(z^2) \right. \\
&\left. - G_k(z)H_k(z)\Phi_x(z)A_k(z^2) \right] z^{-1}dz \quad (25)
\end{aligned}$$

where

$$Q_k(z) = \frac{1}{2}[G_k(z)G_k(z^{-1}) + G_k(-z)G_k(-z^{-1})]. \quad (26)$$

For optimal filterbank design, the analysis $H_k(z)$ and synthesis $G_k(z)$ filters should be chosen so as to minimize the overall signal reconstruction error E given by (23). From Lemma 4 in the Appendix it is immediately seen that, given arbitrary analysis filters, the synthesis filters minimizing (23) are

$$G_k(z) = \frac{H_k(z^{-1})\Phi_x(z)}{2\Phi_{v_k}(z)} = \frac{H_k(z^{-1})\Phi_x(z)}{P_k(z)}, \quad k = 0, 1 \quad (27)$$

where $P_k(z)$ is given by (24). The resulting minimum E is

$$\begin{aligned} 2\pi j E_{\min} &= -\oint \frac{A_0(z^2)H_0(z)H_0(z^{-1})\Phi_x^2(z)z^{-1} dz}{H_0(z)H_0(z^{-1})\Phi_x(z) + H_0(-z)H_0(-z^{-1})\Phi_x(-z)} \\ &\quad - \oint \frac{A_1(z^2)H_1(z)H_1(z^{-1})\Phi_x^2(z)z^{-1} dz}{H_1(z)H_1(z^{-1})\Phi_x(z) + H_1(-z)H_1(-z^{-1})\Phi_x(-z)} \\ &\quad + \oint \Phi_x(z)z^{-1} dz. \end{aligned} \quad (28)$$

It is seen that the optimal $G_k(z)$ are independent of $A_k(z)$. In fact, it may be seen that the expressions (27) under (20), are identical to the perfect reconstruction synthesis filters corresponding to $H_0(z)$, $H_1(z)$. Indeed the conditions for perfect reconstruction [2] are

$$G_k(z)H_k(z) + G_k(-z)H_k(-z) = 2, \quad k = 0, 1. \quad (29)$$

$$G_k(z)H_l(z) + G_k(-z)H_l(-z) = 0, \quad k \neq l, \quad k, l = 0, 1. \quad (30)$$

Since (27) satisfies these, they are the perfect reconstruction synthesis filters corresponding to the analysis filters H_k .

The following proposition has thus been established.

Proposition 1: Given arbitrary analysis filters $H_k(z)$ satisfying (20), the optimal filterbank with Lloyd–Max quantizers is a perfect reconstruction filterbank with synthesis filters given by (27).

This conclusion, establishing perfect reconstruction filterbanks to be optimally structured for the minimization of the overall error variance when Lloyd–Max quantization is used, is a novel result, obtained through the use of the general statistical tools and quantizer models developed in this paper. Although intuitively agreeable, it is by no means obvious, and actually, it specifically holds when Lloyd–Max quantizers are in use. In fact, it is easily seen that the filters produced by minimizing (17) do not in general satisfy the perfect reconstruction conditions (29) and (30) unless (11) and (12) are valid—unless, that is, a Lloyd–Max quantizer described by (8) is used.

We shall now proceed to choose the analysis filters of the perfect reconstruction filterbanks so as to finally minimize the reconstruction error (28). We start by noting that (20) implies the existence of a function $B(z^2)$ such that

$$H_0(z)H_1(z^{-1})\Phi_x(z) = zB(z^2)$$

equivalently

$$H_1(z) = \frac{zH_0(-z^{-1})\Theta(-z^{-1})}{\Theta(z)}B'(z^2) \quad (31)$$

where $B'(z^2)$ is another function of z^2 and

$$\Phi_x(z) = \Theta(z)\Theta(z^{-1})$$

whence

$$H_1(z)H_1(z^{-1})\Phi_x(z) = H_0(-z)H_0(-z^{-1})\Phi_x(-z)[B'(z^2)]^2. \quad (32)$$

Thus,

$$\begin{aligned} H_1(z)H_1(z^{-1})\Phi_x(z) + H_1(-z)H_1(-z^{-1})\Phi_x(-z) \\ = [H_0(z)H_0(z^{-1})\Phi_x(z) + H_0(-z)H_0(-z^{-1}) \\ \times \Phi_x(-z)][B'(z^2)]^2. \end{aligned} \quad (33)$$

Letting $z = e^{j\omega}$ and defining for $k = 0, 1$, the function shown at the bottom of the page, we obtain from (28)

$$\begin{aligned} 2\pi E_{\min} &= \int_{-\pi}^{\pi} \Phi_x(e^{j\omega}) d\omega - \int_{-\pi}^{\pi} A_0(e^{2j\omega})\Phi_x(e^{j\omega})R_0(e^{j\omega}) d\omega \\ &\quad - \int_{-\pi}^{\pi} A_1(e^{2j\omega})\Phi_x(e^{j\omega})R_1(e^{j\omega}) d\omega. \end{aligned}$$

From (32) and (33) $R_1(e^{j\omega}) = R_0(e^{j(\pi+\omega)}) = 1 - R_0(e^{j\omega})$, hence, easily

$$\begin{aligned} 2\pi E_{\min} &= 2 \int_0^{\pi} \Phi_x(e^{j\omega}) d\omega - \int_0^{\pi} [A_0(e^{2j\omega})\Phi_x(e^{j\omega}) \\ &\quad + A_1(e^{2j\omega})\Phi_x(e^{j(\pi+\omega)})]R_0(e^{j\omega}) d\omega \\ &\quad - \int_0^{\pi} [A_0(e^{2j\omega})\Phi_x(e^{j(\pi+\omega)}) \\ &\quad + A_1(e^{2j\omega})\Phi_x(e^{j\omega})]R_1(e^{j\omega}) d\omega. \end{aligned}$$

Since $R_0(e^{j\omega}) + R_1(e^{j\omega}) = 1$, the above is minimized whenever R_0, R_1 assume their maximum values 1 when they respectively multiply the largest of the two above integrands, i.e., $R_0(e^{j\omega}) = 1$ if

$$\begin{aligned} A_0(e^{2j\omega})\Phi_x(e^{j\omega}) + A_1(e^{2j\omega})\Phi_x(e^{j(\pi+\omega)}) \\ > A_0(e^{2j\omega})\Phi_x(e^{j(\pi+\omega)}) + A_1(e^{2j\omega})\Phi_x(e^{j\omega}). \end{aligned}$$

Regrouping terms, this is seen to be equivalent to

$$\Psi(e^{j\omega}) > \Psi(e^{j(\pi+\omega)}), \quad 0 < \pi < \omega$$

where

$$\Psi(e^{j\omega}) = \Phi_x(e^{j\omega})(A_0(e^{2j\omega}) - A_1(e^{2j\omega})). \quad (34)$$

Otherwise, $R_0(e^{j\omega}) = 0$. Equivalently, the minimum error variance is achieved if

$$H_0(e^{j\omega}) = 0 \quad \text{unless } \Psi(e^{j\omega}) > \Psi(e^{j(\pi+\omega)}), \quad 0 < \pi < \omega \quad (35)$$

$$H_1(e^{j\omega}) = 0 \quad \text{unless } \Psi(e^{j\omega}) < \Psi(e^{j(\pi+\omega)}), \quad 0 < \pi < \omega.$$

$$R_k(e^{j\omega}) = \frac{H_k(e^{j\omega})H_k(e^{-j\omega})\Phi_x(e^{j\omega})}{H_k(e^{j\omega})H_k(e^{-j\omega})\Phi_x(e^{j\omega}) + H_k(e^{j(\pi+\omega)})H_k(e^{-j(\pi+\omega)})\Phi_x(e^{j(\pi+\omega)})}$$

The resulting minimum error value is

$$2\pi E_{\min} = \int_{-\pi}^{\pi} \Phi_x(e^{j\omega})(1 - A_1(e^{2j\omega})) d\omega - \int_0^{\pi} \max\{\Psi(e^{j\omega}), \Psi(e^{j(\pi+\omega)})\} d\omega. \quad (36)$$

Note that no assumption whatsoever was made regarding the dependence of $A_0(z)$, $A_1(z)$ on the variances $\sigma_{v_k}^2$ of the quantizer inputs, which are given by

$$\sigma_{v_k}^2 = \frac{1}{2\pi j} \oint (H_k(z)H_k(z^{-1})\Phi_x(z) + H_k(-z)H_k(-z^{-1})\Phi_x(-z))z^{-1} dz.$$

If the form of this dependence were known, a final optimization step would involve the selection of $H_k(z)H_k(z^{-1})\Phi_x(z) + H_k(-z)H_k(-z^{-1})\Phi_x(-z)$ so as to minimize (36).

The corresponding analysis filter $H_1(z)$ in the second band is found from (31) and the synthesis filters from (27). Clearly, the above filters depend explicitly on the statistics of the signal. Since such statistics are often uncertain or unknown, we shall devote the remainder of this Section to the development of a second optimal filter family that is independent of the signal statistics and produces the same minimum error variance E . Precisely as in the case where no quantization exists [10], this family will be developed by minimizing the expression (25) which is equivalent to (23). In fact, directly from Lemma 4 in the Appendix, it is seen that (25) is minimized when

$$H_k(z) = \frac{G_k(z^{-1})}{Q_k(z)}, \quad k = 0, 1 \quad (37)$$

where $Q_k(z)$ is given by (26).

Again, the optimal analysis filters are independent of A_k . The minimization is of course subject to (20); hence, from (37) the synthesis filters must satisfy

$$G_0(z)G_1(z^{-1})\Phi_x(z) + G_0(-z)G_1(-z^{-1})\Phi_x(-z) = 0. \quad (38)$$

Replacing (37) in (25) yields

$$2\pi E_{\min} = \oint \Phi_x(z)z^{-1} dz - \sum_{k=0}^1 \oint \frac{G_k(z)G_k(z^{-1})\Phi_x(z)A_k(z^2)z^{-1} dz}{G_k(z)G_k(z^{-1}) + G_k(-z)G_k(-z^{-1})}$$

whence, easily

$$2\pi E_{\min} = \int_{-\pi}^{\pi} \Phi_x(e^{j\omega}) d\omega - \int_0^{\pi} [A_0(e^{2j\omega}) + A_1(e^{2j\omega})]\Phi_x(e^{j(\pi+\omega)}) d\omega + \sum_{k=0}^1 \int_0^{\pi} S_k(e^{j\omega})A_k(e^{2j\omega})[\Phi_x(e^{j\omega}) - \Phi_x(e^{j(\pi+\omega)})] d\omega$$

where now

$$S_k(e^{j\omega}) = \frac{G_k(e^{j\omega})G_k(e^{-j\omega})}{G_k(e^{j\omega})G_k(e^{-j\omega}) + G_k(e^{j(\pi+\omega)})G_k(e^{-j(\pi+\omega)})}, \quad k=0,1.$$

Clearly, the above will be minimized when $S_0(e^{j\omega})$ and $S_1(e^{j\omega})$ assume their maximum value 1 or minimum value 0. Note, however, that $S_k(e^{j\omega})$ is zero only when $G_k(e^{j\omega})$ is zero. From (38), this implies that if $S_0(e^{j\omega}) = 0$ and $G_0(e^{j\omega})$ is not identically zero, i.e., $G_0(e^{j(\pi+\omega)}) \neq 0$, then necessarily $G_1(e^{j(\pi+\omega)}) = 0$; hence, $S_1(e^{j\omega}) = 1$. Therefore, $S_1(e^{j\omega}) = 1$ for those ω for which $S_0(e^{j\omega}) = 0$. This implies that the above expression will be minimized if $S_0(e^{j\omega}) = 1$ and $S_1(e^{j\omega}) = 0$ whenever

$$A_0(e^{j\omega})[\Phi(e^{j\omega}) - \Phi(e^{j(\pi+\omega)})] > A_1(e^{j\omega})[\Phi(e^{j\omega}) - \Phi(e^{j(\pi+\omega)})].$$

Rearranging terms this leads to

$$\begin{aligned} G_0(z) &= 0, & \text{unless } \Psi(e^{j\omega}) > \Psi(e^{j(\pi+\omega)}) \\ G_1(z) &= 0, & \text{unless } \Psi(e^{j\omega}) < \Psi(e^{j(\pi+\omega)}) \end{aligned} \quad (39)$$

with the analysis filters given by (37). Note that the so derived analysis filter satisfy the statistical decoupling condition (20) because obviously $H_0(z)$ is zero whenever $H_1(z)$ is nonzero and vice versa. For the same reason they satisfy (30). Since (29) is also trivially seen to be satisfied, this implies that the corresponding filterbank has the perfect reconstruction property. As is easily seen, the identical final minimum error variance (36) will then result.

Thus, two optimum filter pairs were obtained using ideal bandreject filters as analysis, respectively synthesis filters. The development is trivially seen to be fully symmetrical with respect to the analysis and synthesis filter choices. More specifically, it is easily seen that (27) may be solved for $H_k(z)$:

$$H_k(z) = \frac{2G_k(z^{-1})\Phi_x^{-1}(z)}{G_k(z)\Phi_x^{-1}(z)G_k(z^{-1}) + G_k(-z)\Phi_x^{-1}(-z)G_k(-z^{-1})}. \quad (40)$$

If the minimum error is then expressed in terms of $G_k(z)$, the optimum $G_k(z)$ will easily be seen to be given by (39). Similarly, solving (37) for $G_k(z)$ yields

$$G_k(z) = \frac{2H_k(z^{-1})}{H_k(z)H_k(z^{-1}) + H_k(-z)H_k(-z^{-1})} \quad (41)$$

and, as before the analysis filters minimizing the corresponding error are easily seen to be given by (35). We summarize the development in this Section in the form of a concluding Theorem.

Theorem 2: Under the restriction (20) of statistically decoupled subbands, the variance E of the overall reconstruction error $\hat{x}[m] - x[m]$ in Fig. 3 is minimized with the use of a perfect reconstruction filterbank. Optimum filter pairs minimizing E are formed by

- 1) analysis filters (35) and synthesis filters (27);
- 2) synthesis filters (39) and analysis filters (40);
- 3) analysis filters (35) and synthesis filters (41);
- 4) synthesis filters (39) and analysis filters (37);

where Ψ is defined by (34). With all pairs the resulting minimum overall error variance is given by (36).

Of the above filter pairs, the last two are independent of the statistics of the input signal, apart from the ordering of the power spectrum values as needed to determine the bandreject regions in (35) and (39). ■

In the specific case where $A_0(e^{2j\omega}) - A_1(e^{2j\omega})$ has a fixed sign for all $\omega \in [0, \pi]$, these filters reduce to those obtained in [10] and [32] and are similar to those obtained in entirely different contexts and for the solution of diverse problems in [17], [18] and [19]. These are defined by the same above expressions with $\Phi_x(e^{j\omega})$ replacing $\Psi(e^{j\omega})$. In particular this will be the case if the transfer functions $A_0(e^{2j\omega})$, $A_1(e^{2j\omega})$ are constants as was assumed in [6] and [8] and as is also indicated to be approximately the case in the experimental results contained in Section VI of the present work.

IV. STRATEGY 2: MINIMIZATION OF THE ERROR COMPONENT UNCORRELATED WITH THE INPUT

Continuing with the second possible strategy, we shall follow the approach in [8] using the above tools and models. As in [6] and [8], the general error expression in (17) may be separated into two terms:

$$2\pi jE = 2\pi jE_F + 2\pi jE_R \quad (42)$$

of which the first E_F is due to the filterbank and depends on the input signal, while the second E_R is due to the noise and is uncorrelated with the input. It is now easily verified that upon selecting $(A_0(z^2)G_0(z), A_1(z^2)G_1(z))$ to be equal to the synthesis filters forming a perfect reconstruction filterbank alongside the analysis filters $(H_0(z), H_1(z))$, the filtering error E_F is set to zero. This of course is to be expected, since if the added noise r is left out of the equivalent filterbank in Fig. 3, the latter will be composed of equivalent synthesis filters equal to $A_0(z^2)G_0(z)$ and $A_1(z^2)G_1(z)$, which will give zero reconstruction error if selected as above.

Once again, to simplify the analysis we shall assume that the analysis filters satisfy (20), in effect decoupling the quantizing operations in each subband. This will imply zero cross-power spectrum $\Phi_{r_0r_1}$ of the noise terms and simplification of (19) to

$$\begin{aligned} 2\pi jE_R = & \oint G_0(z)G_0(z^{-1})A_0(z^2)(1 - A_0(z^2)) \\ & \times \Phi_{v_0}(z)z^{-1} dz + \oint G_1(z)G_1(z^{-1})A_1(z^2) \\ & \times (1 - A_1(z^2))\Phi_{v_1}(z)z^{-1} dz \end{aligned} \quad (43)$$

where (12) was also used. As mentioned above, $A_k(z^2)G_k(z)$ will form a perfect reconstruction bank with $H_k(z)$, hence from (20), (29) and (30)

$$\begin{aligned} & G_k(z)A_k(z^2) \\ &= \frac{2H_k(z^{-1})\Phi_x(z)}{H_k(z)H_k(z^{-1})\Phi_x(z) + H_k(-z)H_k(-z^{-1})\Phi_x(-z)}, \\ & \quad k = 0, 1. \end{aligned} \quad (44)$$

With the above choices for G_0, G_1 the resulting filterbank does not possess the perfect reconstruction property (unless

$A_k = 1$), but the filterbank error E_F becomes zero, leaving only the random error E_R to impede perfect reconstruction. From (43) and (44), the latter is seen to become

$$\begin{aligned} 2\pi jE_R = & \sum_{k=0}^1 \oint \frac{1 - A_k(z^2)}{A_k(z^2)} \\ & \times \frac{H_k(z)H_k(z^{-1})\Phi_x^2(z)z^{-1} dz}{H_k(z)H_k(z^{-1})\Phi_x(z) + H_k(-z)H_k(-z^{-1})\Phi_x(-z)}. \end{aligned} \quad (45)$$

This is identical to the first two terms of (28) with the factors $A_k(z^2)$ replaced by $-(1 - A_k(z^2))/A_k(z^2)$. Thus, as before, this is minimized if $H_0(e^{j\omega}) = 0$ unless $\Psi(e^{j\omega}) > \Psi(e^{j(\pi+\omega)})$ where now

$$\Psi(e^{j\omega}) = \left(\frac{1}{A_1(e^{2j\omega})} - \frac{1}{A_0(e^{2j\omega})} \right) \Phi_x(e^{j\omega}) \quad (46)$$

and the minimum error is

$$\begin{aligned} 2\pi E_R = & \int_{-\pi}^{\pi} \Phi_x(e^{j\omega}) \frac{1 - A_1(e^{2j\omega})}{A_1(e^{2j\omega})} d\omega \\ & - \int_0^{\pi} \max\{\Psi(e^{j\omega}), \Psi(e^{j(\pi+\omega)})\} d\omega. \end{aligned} \quad (47)$$

As with strategy 1, the same minimum error E_R may be obtained using synthesis filters independent from the input signal statistics. To see this, replace $\Phi_{v_k}(z)$ from (6) in (43), change integration variables, and rearrange terms to obtain the equivalent expression

$$\begin{aligned} 2\pi jE_R = & \sum_{k=0}^1 \oint A_k(z^2)(1 - A_k(z^2)) \\ & \times H_k(z)H_k(z^{-1}) \frac{1}{2} [G_k(z)G_k(z^{-1}) \\ & + G_k(-z)G_k(z^{-1})] \Phi_x(z)z^{-1} dz. \end{aligned}$$

Thus, if

$$\begin{aligned} H_k(z) = & \frac{2G_k(z^{-1})A_k(z^2)}{A_k^2(z^2)[G_k(z)G_k(z^{-1}) + G_k(-z)G_k(-z^{-1})]} \\ = & \frac{2G_k(z^{-1})/A_k(z^2)}{G_k(z)G_k(z^{-1}) + G_k(-z)G_k(-z^{-1})} \end{aligned} \quad (48)$$

the random error variance becomes

$$\begin{aligned} 2\pi jE_R = & \sum_{k=0}^1 \oint \frac{1 - A_k(z^2)}{A_k(z^2)} \\ & \times \frac{G_k(z)G_k(z^{-1})\Phi_x(z)z^{-1} dz}{G_k(z)G_k(z^{-1}) + G_k(-z)G_k(-z^{-1})} \end{aligned}$$

which is minimized, precisely as in the preceding Section, with synthesis filters (39) with $\Psi(e^{j\omega})$ now given by (46). The filterbank composed of analysis filters (49) and equivalent synthesis filters $G_k(z)A_k(z^2)$ is trivially seen to have the perfect reconstruction property, hence to render the filtering error E_F equal to zero. The resulting minimum value of E_R

is identical to (47). The same minimum value of E_R with $E_F = 0$ is found with expressions symmetric to (44) and (48).

$$H_k(z) = \frac{1}{A_k(z^2)} \times \frac{2G_k(z^{-1})\Phi_x^{-1}(z)}{G_k(z)\Phi_x^{-1}(z)G_k(z^{-1}) + G_k(-z)\Phi_x^{-1}(-z)G_k(-z^{-1})} \quad (49)$$

with $G_k(z)$ given by (39), and with

$$G_k(z)A_k(z^2) = \frac{2H_k(z^{-1})}{H_k(z)H_k(z^{-1}) + H_k(-z)H_k(-z^{-1})} \quad (50)$$

with $H_k(z)$ given by (35).

The results of this Section are summarized in the following theorem.

Theorem 3: Under the assumption (20) of statistically decoupled subbands, the variance E_F or the error correlated with the input becomes zero and the variance E_R of the random error uncorrelated with the input is minimized with the use of a nonperfect-reconstruction filterbank, formed with any one of the following filter choices:

- 1) analysis filters (35) and synthesis filters (44);
- 2) synthesis filters (39) and analysis filters (49);
- 3) analysis filters (35) and synthesis filters (50);
- 4) synthesis filters (39) and analysis filters (48);

where Ψ is now defined by (46). With any one choice, the minimum error variance E_R is given by (47). ■

In the special case where $A_0(e^{2j\omega}) - A_1(e^{2j\omega})$ has a fixed sign for all $\omega \in [0, \pi]$, these filters are identical to the ideal filters minimizing the overall error (17).

A final minimization of the resulting random error (47) may be achieved by selecting the magnitude of the bandreject filters in their passband so as to minimize (47). Note that apart from this, no further reduction of the random error variance is possible by use of linear postfiltering, since such postfiltering may equivalently be incorporated in the subband filterbank that is being optimized.

V. GENERALIZATION

The approach in the preceding Sections is completely general and is easily extended to the case where the input is a multidimensional signal of dimension N and the filterbank has M channels. In the general case, if $\mathbf{z} = [z_1, \dots, z_N]^T$, $\underline{\omega} = [\omega_1, \dots, \omega_N]^T$ and if $\mathbf{M}$ is the sampling matrix with

$$M = \det \mathbf{M} \quad (51)$$

the power spectral densities (Fig. 3) are related by

$$\Phi_{v_k x}(e^{j\underline{\omega}}) = \frac{1}{M} H_k(e^{j\underline{\omega}}) \Phi_x(e^{j\underline{\omega}}) \quad (52)$$

$$\begin{aligned} \Phi_{v_k v_l}(e^{j\underline{\omega}}) &= \frac{1}{M^2} \sum_{i=0}^{M-1} H_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) \\ &\times \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) H_l(e^{-j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) \end{aligned} \quad (53)$$

where the notation in [2], [12], [13], is used, $\mathbf{q}_i$ are the coset vectors of $\mathbf{M}^T$ and $\mathbf{M}^{-T}$ is a short notation for $(\mathbf{M}^T)^{-1}$. From the above, the condition for statistical decoupling of the subbands is

$$\begin{aligned} \sum_{i=0}^{M-1} H_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) \\ \times H_l(e^{-j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) = 0 \quad k \neq l. \end{aligned} \quad (54)$$

As before, the noise spectra are

$$\Phi_{r_k}(e^{j\underline{\omega}}) = A_k(e^{j\mathbf{M}^T \underline{\omega}}) (1 - A_k(e^{j\mathbf{M}^T \underline{\omega}})) \Phi_{v_k}(e^{j\underline{\omega}}) \quad (55)$$

and

$$\Phi_{y_k x}(e^{j\underline{\omega}}) = \frac{1}{M} G_k(e^{j\underline{\omega}}) A_k(e^{j\mathbf{M}^T \underline{\omega}}) H_k(e^{j\underline{\omega}}) \Phi_x(e^{j\underline{\omega}}) \quad (56)$$

$$\begin{aligned} \Phi_{y_k}(e^{j\underline{\omega}}) &= G_k(e^{j\underline{\omega}}) (A_k^2(e^{j\mathbf{M}^T \underline{\omega}}) \Phi_{v_k}(e^{j\underline{\omega}}) \\ &+ \Phi_{r_k}(e^{j\underline{\omega}})) G_k(e^{-j\underline{\omega}}). \end{aligned} \quad (57)$$

The error variance to be minimized is

$$E = E \left\{ \left(\sum_{k=0}^{M-1} y_k[m] - x[m] \right)^2 \right\}.$$

Thus, since from (54), $E\{y_k[m]y_l[m]\} = 0$ if $k \neq l$

$$E = E \left\{ \sum_{k=0}^{M-1} y_k^2[m] - 2 \sum_{k=0}^{M-1} y_k[m]x[m] + x^2[m] \right\}$$

and, using (56) and (57)

$$\begin{aligned} (2\pi)^N E &= \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} G_k(e^{j\underline{\omega}}) G_k(e^{-j\underline{\omega}}) A_k^2(e^{j\mathbf{M}^T \underline{\omega}}) \Phi_{v_k}(e^{j\underline{\omega}}) d\underline{\omega} \\ &+ \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} G_k(e^{j\underline{\omega}}) G_k(e^{-j\underline{\omega}}) \Phi_{r_k}(e^{j\underline{\omega}}) d\underline{\omega} \\ &- \frac{2}{M} \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} G_k(e^{j\underline{\omega}}) H_k(e^{j\underline{\omega}}) A_k(e^{j\mathbf{M}^T \underline{\omega}}) \\ &\times \Phi_x(e^{j\underline{\omega}}) d\underline{\omega} + \int_{-\pi}^{\pi} \Phi_x(e^{j\underline{\omega}}) d\underline{\omega}. \end{aligned}$$

Using (53) and (55), the above simplifies to

$$\begin{aligned} (2\pi)^N E &= \int_{-\pi}^{\pi} \Phi_x(e^{j\underline{\omega}}) d\underline{\omega} + \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} \frac{1}{M} G_k(e^{j\underline{\omega}}) G_k(e^{-j\underline{\omega}}) \\ &\times P_k(e^{j\underline{\omega}}) A_k(e^{j\mathbf{M}^T \underline{\omega}}) d\underline{\omega} - \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} \frac{2}{M} H_k(e^{j\underline{\omega}}) \\ &\times G_k(e^{j\underline{\omega}}) \Phi_x(e^{j\underline{\omega}}) A_k(e^{j\mathbf{M}^T \underline{\omega}}) d\underline{\omega} \end{aligned} \quad (58)$$

where

$$\begin{aligned} P_k(e^{j\underline{\omega}}) &= \frac{1}{M} \sum_{i=0}^{M-1} H_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) \\ &\times \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}) H_k(e^{-j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_i)}). \end{aligned} \quad (59)$$

Replacing (59) in (58) regrouping terms and changing variables, we easily obtain the equivalent expression

$$(2\pi)^N E = \int_{-\pi}^{\pi} \Phi_x(e^{j\omega}) d\omega + \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} \frac{1}{M} H_k(e^{j\omega}) \times H_k(e^{-j\omega}) Q_k(e^{j\omega}) A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j\omega}) d\omega - \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} \frac{2}{M} H_k(e^{j\omega}) G_k(e^{j\omega}) \times A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j\omega}) d\omega \quad (60)$$

where

$$Q_k(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} G_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) \times G_k(e^{-j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}), \quad (61)$$

In the above, $d\omega$ is a short notation for the multiple differential $d\omega_1 \cdots d\omega_N$, and multiple integration is understood to be denoted by the integration sign. As discussed earlier, two available strategies are defined by

- 1) minimizing the overall reconstruction error;
- 2) setting to zero the signal-related component of the reconstruction error and minimizing the residual random error.

The first involves minimizing (58). As in Section III, using Lemma 4 in the Appendix, this is seen to be accomplished with a first optimal filter bank with synthesis filters

$$G_k(e^{j\omega}) = \frac{H_k(e^{-j\omega}) \Phi_x(e^{j\omega})}{P_k(e^{j\omega})} \quad (62)$$

where $P_k(e^{j\omega})$ is given by (59). From (54) and (62)

$$\sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) G_l(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = 0 \quad k \neq l$$

$$\sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) G_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = M P_k(e^{j\omega}). \quad (63)$$

As shown in Lemma 5 in the Appendix, these define a perfect reconstruction filterbank with analysis filters H_k and synthesis filters G_k . As also shown in the same Lemma, the analysis filters are arbitrary filters such that

$$H_k(e^{j\omega}) = \Theta(e^{j\omega}) \mathbf{L}_k(e^{j\omega})$$

where $\Theta(z)$ is defined by

$$\Phi_x(e^{j\omega}) = \Theta(e^{j\omega}) \Theta(e^{-j\omega})$$

and the polyphase matrix of the analysis filterbank $H_k(e^{j\omega})$ satisfies

$$\mathbf{F}_p^T(e^{-j\omega}) \mathbf{F}_p(e^{j\omega}) = \text{diag}\{P_i(e^{j\omega})\}_i.$$

The resulting minimum error is

$$(2\pi)^N E_{\min} = \int_{-\pi}^{\pi} \Phi_x(e^{j\omega}) d\omega - \frac{1}{M} \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} G_k(e^{j\omega}) \times H_k(e^{j\omega}) A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j\omega}) d\omega = \int_{-\pi}^{\pi} \Phi_x(e^{j\omega}) d\omega - \frac{1}{M} \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} A_k(e^{j\mathbf{M}^T \omega}) \times \Phi_x(e^{j\omega}) \frac{H_k(e^{j\omega}) \Phi_x(e^{j\omega}) H_k(e^{-j\omega})}{P_k(e^{j\omega})} d\omega. \quad (64)$$

If all $A_k = 1$, relation (A18) in Lemma 5 in the Appendix shows that the minimum error will reduce to zero. This is, of course, expected, since a perfect reconstruction filterbank is being used. If $A_k \neq 1$, the error (64) will be minimized under conditions specified by Lemma 6 in the Appendix (see also [22]). Specifically, if

$$\Omega = \{\omega : \mathbf{v} = \mathbf{M}^T \omega, \quad -\pi < v_i < \pi\} \quad (65)$$

application of Lemma 6 in the Appendix with

$$\phi_k(\omega) = A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j\omega})$$

$$R_k(e^{j\omega}) = \frac{H_k(e^{j\omega}) \Phi_x(e^{j\omega}) H_k(e^{-j\omega})}{M P_k(e^{j\omega})}$$

shows that minimization of (64) is achieved as follows. Let Ω_i be the set of $\omega \in \Omega$ for which the permutation $\{p_i(k)\}$ of $\{0, 1, \dots, M-1\}$ maximizes the sum

$$\max_{\{p_i(k)\}} \sum_{k=0}^{M-1} A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{p_i(k)})})$$

i.e., Ω_i is defined as the set of $\omega \in \Omega$ such that

$$\sum_{k=0}^{M-1} A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{p_i[k])})}) \geq \sum_{k=0}^{M-1} A_k(e^{j\mathbf{M}^T \omega}) \Phi_x(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{p'_i[k])})}) \quad (66)$$

for all permutations $\{p'_i(k)\}$ of $\{0, 1, \dots, M-1\}$. Then the analysis filters minimizing (64) are defined by

$$R_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{p_i(k)})}) = \begin{cases} 1, & \omega \in \Omega_i \\ 0, & \text{elsewhere} \end{cases}.$$

Equivalently

$$H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{p_i(k)})}) = 0 \quad \text{unless } \omega \in \Omega_i. \quad (67)$$

Similarly, if the second strategy is pursued, of setting to zero the error component which is correlated with the signal, the remaining error term in the general case is

$$(2\pi)^N E_R = \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} G_k(e^{j\omega}) G_k(e^{-j\omega}) \Phi_{r_k}(e^{j\omega}) d\omega \quad (68)$$

where as before

$$G_k(e^{j\omega})A_k(e^{jM^T\omega}) = \frac{H_k(e^{-j\omega})\Phi_x(e^{j\omega})}{P_k(e^{j\omega})} \quad (69)$$

with $P_k(e^{j\omega})$ defined in (59). On replacing in (68) and using (55) we find

$$(2\pi)^N E_R = \frac{1}{M} \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} \frac{1 - A_k(e^{jM^T\omega})}{A_k(e^{jM^T\omega})} \times \Phi_x(e^{j\omega}) \frac{H_k(e^{j\omega})\Phi_x(e^{j\omega})H_k(e^{-j\omega})}{P_k(e^{j\omega})} d\omega \quad (70)$$

which differs from the second term in (64) only in having the factor A_k replaced by $-(1 - A_k)/A_k$. The same, therefore, development will give optimal analysis filters identical to (67) but with Ω_i defined by

$$\max_{\{p_i(k)\}} \sum_{k=0}^{M-1} \frac{-1 + A_k(e^{jM^T\omega})}{A_k(e^{jM^T\omega})} \Phi_x(e^{j(\omega+2\pi M^{-T}q_{p_i(k)})}). \quad (71)$$

Repeating the procedure detailed in Sections III and IV for the minimization of (60), yields optimal filters consisting of

$$H_k(e^{j\omega}) = \frac{G_k(e^{-j\omega})}{Q_k(e^{j\omega})}, \quad k = 0, \dots, M-1 \quad (72)$$

where $Q_k(e^{j\omega})$ is given by (61). The optimal synthesis filters in this family are found as before to be

$$G_k(e^{j(\omega+2\pi M^{-T}q_{p_i(k)})}) = 0 \quad \text{unless } \omega \in \Omega_i \quad (73)$$

where Ω_i is defined by (66) for the first strategy and (71) for the second. The symmetric filter expressions of (74), shown at the bottom of the page, with G_k given by (73), and

$$G_k(e^{j\omega}) = \frac{MH_k(e^{-j\omega})}{\sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi M^{-T}q_i)})H_k(e^{j(\omega+2\pi M^{-T}q_i)})} \quad (75)$$

with H_k given by (67), give the same minimum overall error. The same filters with G_k replaced by $A_k G_k$ also set E_F to zero and minimize the variance of E_R . We summarize the above results in the form of the following final Theorem.

Theorem 4: In the general multidimensional case, the variance E of the overall reconstruction error is minimized with the use of a perfect reconstruction filter bank with either of the following filter choices:

- 1) analysis filters (67) and synthesis filters (62);
- 2) synthesis filters (73) and analysis filters (74);
- 3) analysis filters (67) and synthesis filters (75);
- 4) synthesis filters (73) and analysis filters (72);

where Ω_i is defined by (66). The same filter bank expressions with G_k replaced by $A_k G_k$ and Ω_i defined by (71) describe a nonperfect reconstruction filterbank setting to zero the variance E_F of the error correlated with the input and minimizing the variance E_R of the random error uncorrelated with the input. ■

VI. EXPERIMENTAL APPLICATION TO IMAGE CODING

In all of the preceding discussion, particular emphasis was placed on rigor and generality so as to construct a coherent theoretical tool for the analysis and minimization of the quantization effects in subband filterbanks with Lloyd–Max quantizers. In approaching the specific application of image coding, the same level of rigor cannot be sustained. Several practical problems arise, including the following.

- The development of vector Lloyd–Max quantizers with K large enough to ensure fully uncorrelated sequences $u[m]$ and $r[m]$ is impractical.
- Optimal scalar coders for images described by generalized Gaussian statistics were developed in [24] and used in [6] and [8]. However, there has been no attempt to generalize these coders for the construction of optimal vector codebooks for images with similar statistics.

The above difficulties are due to the scarcity of reliable multivariate statistical descriptions for broad classes of images. A codebook based on multivariate histograms of individual images, would vary from image to image and its transmission would command unacceptable overhead in bit rate.

Optimal vector Lloyd–Max quantizers with general applicability may easily be formed for images with Gaussian statistics, zero mean and unit variance, and known horizontal and vertical correlation coefficients μ_1, μ_2 . Samples of such images may be generated by the autoregressive moving average (ARMA) model [23]

$$\begin{aligned} x[m_1, m_2] = & \mu_1 x[m_1 - 1, m_2] + \mu_2 x[m_1, m_2 - 1] \\ & - \mu_1 \mu_2 x[m_1, m_2 - 1] \\ & + \sqrt{(1 - \mu_1^2)(1 - \mu_2^2)} \epsilon[m_1, m_2] \end{aligned} \quad (76)$$

where $\epsilon[m_1, m_2]$ is a Gaussian, zero-mean, unit-variance, white sequence. In subband coding, this method will be useful for the representation of the top (lowpass) pyramid image, which may be assumed to be well described by Gaussian statistics. It is much less useful, for the remaining images, which have been shown to be best described by generalized Gaussian statistics with $c = 0.5$ [25].

The 256×256 image Lenna was used to test the implementation of these methods. This is an image with an autocorrelation sequence adequately approximated by

$$R_x[p_1, p_2] = T \mu_1^{|p_1|} \mu_2^{|p_2|}$$

and thus, with approximate power spectral density

$$\begin{aligned} \Phi_x(z_1, z_2) \\ = \frac{T^2(1 - \mu_1^2)(1 - \mu_2^2)}{(1 - \mu_1 z_1)(1 - \mu_1 z_1^{-1})(1 - \mu_2 z_2)(1 - \mu_2 z_2^{-1})}. \end{aligned} \quad (77)$$

A number of tests were made, whose preliminary results were reported in [21] and [24]. In the first test, in order to

$$H_k(e^{j\omega}) = \frac{MG_k(e^{-j\omega})\Phi_x^{-1}(e^{j\omega})}{\sum_{i=0}^{M-1} G_k(e^{j(\omega+2\pi M^{-T}q_i)})\Phi_x^{-1}(e^{j(\omega+2\pi M^{-T}q_i)})G_k(e^{-j(\omega+2\pi M^{-T}q_i)})} \quad (74)$$

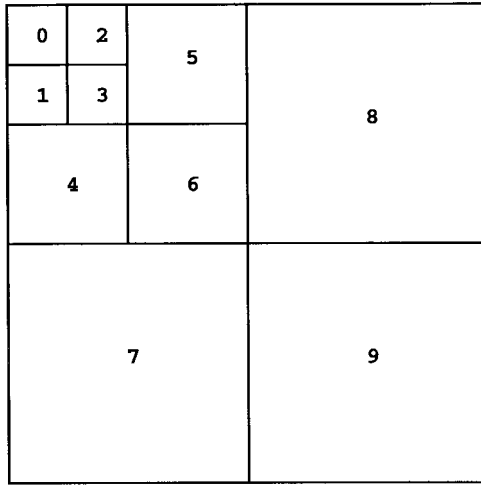


Fig. 4. Subband split for a three-level pyramid.

TABLE I

 GAIN FACTORS OF THE MODEL FOR THE VECTOR QUANTIZER FOR THE CODING AT 5 B/PIXEL OF THE 32×32 LOWPASS SUBBAND

Subband	a_{00}	a_{10}	a_{01}
0	0.999726	0.000102	0.000217

construct the general model (8), the optimal vector Lloyd–Max quantizer for the coding of the top (lowpass, image 0 in Fig. 4) pyramid image was designed as follows. Equation (76) with $\mu_1 = \mu_2 = 0.9$ was repeatedly used to generate the images for the training set. For this class of images, the optimal codebook was obtained for vectors composed of the pixel intensities in a 3×1 image block. The K -means algorithm was used to generate a vector codebook, which approximates the optimal vector Lloyd–Max quantizer [15]. A transfer function of the form

$$A(z) = a_0 + a_1(z + z^{-1}) + a_2(z^2 + z^{-2})$$

was then assumed, for the realization of (8) and the coefficients a_{kl} were obtained by using (A13) in the Appendix, which in this case translates to

$$R_{\tilde{s}\tilde{v}}[p] = \sum_{k=0}^2 a_k (R_{\tilde{v}}[p-k] + R_{\tilde{v}}[p+k]).$$

Least-squares-fit techniques were used for the evaluation of a_{kl} for the top (lowpass), 32×32 pyramid image produced by the analysis filter in Table III, which is a good approximation of an ideal lowpass filter in $[0, \pi/2]$. The results shown in Table I indicate that $|a_{10}|, |a_{01}|$ were always of the order of 1% of a_{00} and hence, that the function $A(z)$ may credibly be assumed to be constant.

In a second test, the optimal scalar Lloyd–Max quantizers developed in [6] were used. These quantizers assume generalized Gaussian densities

$$f(x) = ae^{-\beta|x|^c} \quad (78)$$

with $c = 0.75$ for the top lowpass image (image 0 in Fig. 4) and $c = 0.5$ for the remaining images (images 1

 TABLE II
GAIN FACTORS OF THE MODEL FOR THE LLOYD–MAX QUANTIZER FOR 0.5 B/PIXEL, 0.7 B/PIXEL, AND 1.5 B/PIXEL

Subband	0.5 bits/pixel		0.7 bit/pixel		1.5 bit/pixel	
	a_0	a_1	a_0	a_1	a_0	a_1
0	0.999827	0.000251	0.999827	0.000251	0.999827	0.000251
1	0.990525	0.018047	0.990525	0.018047	0.997136	-0.023244
2	0.974025	-0.002037	0.974025	-0.002037	0.997218	0.005852
3	0.973516	0.0047235	0.973516	0.0047235	1.009581	-0.018853
4	0.925825	-0.058862	0.925825	-0.058862	0.991447	-0.029553
5	0.846875	-0.005443	0.893652	-0.012763	0.958066	-0.017361
6	0.853849	-0.016818	0.853849	-0.016818	0.973114	-0.011451
7	0.0	0.0	0.685886	0.0751473	0.904878	-0.034062
8	0.0	0.0	0.0	0.0	0.632778	-0.024948
9	0.0	0.0	0.0	0.0	0.651410	0.042759

TABLE III

LOWPASS ANALYSIS FILTER COEFFICIENTS

Filter Coefficient $h_0[n]$	$ n $
0.5	0
0.3056509920387	1
-0.089954382362	3
0.0434268789585	5
-0.022368859346	7
0.0109640072736	9
-0.0048571826986	11
0.00220495962112	13
-0.0019109649757	15

through 9 in Fig. 4). These images were obtained using the quadrature mirror filters (QMF's) in [6]. Again, the resulting cross and autocorrelation functions were evaluated setting now for greater simplicity $A(z) = a_0 + a_1(z + z^{-1})$ and using again (A13) in the Appendix for the evaluation of a_0, a_1 . The results, summarized in Table II, suggest that again the assumption of constant $A(z) = a_0$ is highly credible, since the value of the coefficients $|a_k|$ with $k \neq 0$ does not rise above 10% of the value of a_0 . The vector and scalar Lloyd–Max quantizers for the coding of the top (lowpass) 32×32 image were then compared. It was seen that for this subband the vector quantizer performed much better than the scalar quantizer, giving an equal-quality (50 dB) reconstruction of the lowpass subband, at a savings of 1.5 b/pixel. However, the size of this image is such that this savings does not translate to a significant overall bit rate decrease. For the remaining bands, the correlation coefficients of adjacent pixels was found to be less than 0.1, permitting the assumption that for these subbands the optimal vector Lloyd–Max quantizer is well approximated by the scalar Lloyd–Max quantizer. This assumption is supported by the results in Table II.

These results also support the assumption tacitly made in [6]–[8], that the quantizer model (8) with a simple constant transfer function will produce reasonably uncorrelated sequences of quantization input and quantization noise.

To apply the results of Section V in this case, we shall start with a single division into four bands by the $\mathbf{M} = \text{diag}\{2, 2\}$ sampling matrix. Since the functions $A_k(z^2)$ are constants a_k ,

the quantity to be maximized is

$$I = \sum_{k=0}^3 a_k \int_{\Omega} \left[\sum_{l=0}^3 \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)}) \times R_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)}) \right] d\underline{\omega} \quad (79)$$

where $\mathbf{q}_0 = [0 \ 0]^T$, $\mathbf{q}_1 = [1 \ 0]^T$, $\mathbf{q}_2 = [0 \ 1]^T$, $\mathbf{q}_3 = [1 \ 1]^T$. We shall assume nonincreasing a_k :

$$a_0 \geq a_1 \geq a_2 \geq a_3 \quad (80)$$

which is a reasonable assumption, since experience shows that for best performance, the lowpass-lowpass filter quantization should be finer, the highpass-highpass filter quantization coarser, and that of the remaining two approximately equally coarse. Now, as is easily verified (for a formal proof see, e.g., [26, Lemma 5.1, p. 126]), in order to maximize (79), the integrals multiplying the a_k in (79) should also form a correspondingly nonincreasing sequence, i.e.,

$$\begin{aligned} & \sum_{l=0}^3 \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)}) R_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)}) \\ & \geq \sum_{l=0}^3 \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)}) R_{k-1}(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)}). \end{aligned}$$

Since from Lemma 6 in the Appendix, each of these integrals will contain a single term $R_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_l)})$ equal to one with the rest being zero, this means that (66) may be more simply expressed as follows: for each permutation $p_i[k]$ of $k = 0, \dots, M-1$, $i = 1, \dots, M!$, let Ω_i be the set of $\underline{\omega} \in \Omega$ for which

$$\Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_{p_i[k]})}) \geq \Phi_x(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_{p_i[k-1]})}). \quad (81)$$

Then, the analysis filters minimizing (79), satisfy

$$H_k(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_{p_i[k]})}) = 0 \quad \text{unless } \underline{\omega} \in \Omega_i. \quad (82)$$

Note also that since the sequence $\frac{(a_k-1)}{a_k}$ in (71) is also nonincreasing

$$\frac{(a_0-1)}{a_0} \geq \frac{(a_1-1)}{a_1} \geq \frac{(a_2-1)}{a_2} \geq \frac{(a_3-1)}{a_3}. \quad (83)$$

The same family of filters will also minimize (71) and hence will also optimize the second strategy analyzed in Section IV. To find the optimal filters, we shall assume an image spectral density of the form (77) with $\mu_1 = \mu_2 = \mu$:

$$\Phi_x(e^{j\underline{\omega}}) = \frac{T^2(1-\mu^2)}{(1+\mu^2-2\mu\cos\omega_1)(1+\mu^2-2\mu\cos\omega_2)}. \quad (84)$$

In this case, $\Omega = \{|\omega_i| < \pi/2, i = 1, 2\}$ and $\mathbf{q}_0 = [0 \ 0]^T$, $\mathbf{q}_1 = [0 \ 1]^T$, $\mathbf{q}_2 = [1 \ 0]^T$, $\mathbf{q}_3 = [1 \ 1]^T$. As is easily seen then, since on Ω the inequality $\cos\omega_1 > \cos\omega_2$ is equivalent to $|\omega_1| < |\omega_2|$, if (80) is strictly monotonic, inequality (81) will be valid for the permutation $p_1 = \{0, 1, 2, 3\}$ on

$$\Omega_1 = \{\underline{\omega} \in \Omega : |\omega_1| < |\omega_2|\}. \quad (85)$$

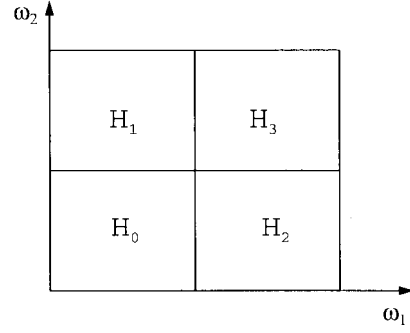


Fig. 5. Supports of the optimal analysis filters.

TABLE IV
OPTIMAL BIT ALLOCATION AMONG SUBBANDS FOR
BIT RATES 0.5 B/PIXEL, 0.7 B/PIXEL, AND 1.5 B/PIXEL

Subband	0.5 bpp	0.7 bpp	1.5 bpp
0	128	128	128
1	31	31	63
2	15	15	63
3	15	15	31
4	7	7	31
5	5	7	15
6	5	5	7
7	-	3	7
8	-	-	3
9	-	-	3

It will also be valid for the permutation $p_2 = \{0, 2, 1, 3\}$ on

$$\Omega_2 = \{\underline{\omega} \in \Omega : |\omega_1| > |\omega_2|\}. \quad (86)$$

Obviously, $\Omega = \Omega_1 \cup \Omega_2$ and all other permutations p_i , $i = 3, \dots, 24$, lead to empty corresponding sets Ω_i . The optimal filters are defined by (81) and (82) as follows:

$$H_0(e^{j\underline{\omega}}) = 0, \quad \text{unless } \underline{\omega} \in \Omega \quad (87)$$

$$H_1(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_1)}) = H_1(e^{j[\omega_1 \ \omega_2 + \pi]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega_1$$

$$H_1(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_2)}) = H_1(e^{j[\omega_1 + \pi \ \omega_2]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega_2$$

and

$$H_2(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_1)}) = H_2(e^{j[\omega_1 \ \omega_2 + \pi]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega_2$$

$$H_2(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_2)}) = H_2(e^{j[\omega_1 + \pi \ \omega_2]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega_1.$$

Finally,

$$H_3(e^{j(\underline{\omega} + 2\pi \mathbf{M}^{-T} \mathbf{q}_3)}) = H_3(e^{j[\omega_1 + \pi \ \omega_2 + \pi]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega. \quad (88)$$

This may be seen to produce a complicated filterbank requiring the realization of filters with diamond- and fan-shaped support. However, since subbands 1 and 2 are approximately equally

TABLE V
VARIANCE OF THE RECONSTRUCTION ERROR FOR THE HAAR, D2, AND D6, THE FILTER PAIRS ACHIEVING BEST AND WORST PSNR AMONG THOSE TESTED IN [27], AND THE OPTIMAL FILTERS FOR BIT RATES OF 0.5 B/PIXEL, 0.7 B/PIXEL, AND 1.5 B/PIXEL

Filter	0.5 bits/pixel		0.7 bits/pixel		1.5 bits/pixel	
	strategy 1	strategy 2	strategy 1	strategy 2	strategy 1	strategy 2
Haar	132.31	135.49	91.72	102.93	28.65	33.54
D2	95.08	96.61	64.92	70.05	20.89	24.38
D6	79.32	83.73	55.02	63.34	18.36	22.20
Best PSNR in [27]	76.36	79.89	51.63	60.28	15.77	19.37
Worst PSNR in [27]	80.68	83.87	55.91	64.02	16.78	20.72
Optimal	70.95	73.47	46.27	56.68	14.72	18.37

coarsely quantized, hence $a_1 \cong a_2$, approximately the same value of I in (79) will be obtained with the simpler filters

$$H_1(e^{j(\underline{\omega}+2\pi\mathbf{M}^{-T}\mathbf{q}_1)}) = H_1(e^{j[\omega_1+\pi]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega \quad (89)$$

$$H_2(e^{j(\underline{\omega}+2\pi\mathbf{M}^{-T}\mathbf{q}_2)}) = H_2(e^{j[\omega_1+\pi, \omega_2]^T}) = 0, \quad \text{unless } \underline{\omega} \in \Omega. \quad (90)$$

The supports of these filters are shown in Fig. 5.

A three-level subband-split pyramid was generated for the representation of this image (see Fig. 4). Family (i) of the optimal filters was first implemented. An approximation of an ideal lowpass filter with passband $[0, \pi/2]$ was used as analysis filter of the primary band. This is a symmetric 31-tap finite impulse response (FIR) filter with coefficients $h_0[2k] = 0$, $k \neq 0$, and $h_0[0]$, and $h_0[2k+1]$ given in Table III. All other filters of the analysis/synthesis quadruple were defined from the choice of the filter $H_0(z)$ using (27), (31) with $B'(z^2)$ selected so as to whiten the output of the highpass filter. Specifically, with $H_0(z)$ selected as above, the remaining filters were found by (27) and

$$H_1(z) = \frac{zH_0(-z^{-1})}{(1+\mu)\Phi_x(z)M(z^2)} \quad (91)$$

$$G_1(z) = (1+\mu)^2 H_1(z^{-1})\Phi_x(z) = \frac{(1+\mu)z^{-1}H_0(-z)}{M(z^{-2})} \quad (92)$$

where

$$\frac{H_0(z)H_0(z^{-1})}{\Phi_x(-z)} + \frac{H_0(-z)H_0(-z^{-1})}{\Phi_x(z)} = 2M(z^2)M(z^{-2}). \quad (93)$$

With these choices it is seen that the highpass output is

$$H_1(z)\Phi_x(z)H_1(z^{-1}) + H_1(-z)\Phi_x(z)H_1(-z^{-1}) = \frac{2}{(1+\mu)^2}.$$

The above filters are applied first to lines then to columns of the image. This is clearly equivalent to a four subband filter bank with filters

$$\begin{aligned} H_0(z_1, z_2) &= H_0(z_1)H_0(z_2), & G_0(z_1, z_2) &= G_0(z_1)G_0(z_2) \\ H_1(z_1, z_2) &= H_0(z_1)H_1(z_2), & G_1(z_1, z_2) &= G_0(z_1)G_1(z_2) \\ H_2(z_1, z_2) &= H_1(z_2)H_2(z_2), & G_2(z_1, z_2) &= G_1(z_2)G_2(z_2) \\ H_3(z_1, z_2) &= H_1(z_1)H_1(z_2), & G_3(z_1, z_2) &= G_1(z_1)G_1(z_2). \end{aligned}$$

It is trivially seen that the resulting H_0, H_1, H_2, H_3 approximate the ideal bandreject filters in (87), (89), and (90). It is also seen that the resulting $G_k(z_1, z_2)$, $k = 0, 1, 2, 3$, satisfy (62) and (64).

The optimal filters supports are easily seen to remain the same in each of the repeated decompositions of the lowpass image to form the final decomposition. Note however, that the spectral densities Φ_x change in each step of the decomposition. Specifically, while in the first step of the decomposition $\Phi_x^{(0)}(z)$ is given by (77), in the remaining steps it is calculated by

$$\begin{aligned} \Phi_x^{(i+1)}(z) &= [H_0(z)\Phi_x^{(i)}(z)H_0(z^{-1}) \\ &\quad + H_0(-z)\Phi_x^{(i)}(-z)H_0(-z^{-1})]_{z \rightarrow \sqrt{z}} \end{aligned}$$

for $i = 0, 1, 2$. To avoid explicit dependence on the filter statistics, a filter bank using family (iv) of the optimal filters in Theorem 4 was also implemented using $G_0(z)$ equal to the 31-tap FIR filter in Table III

$$G_1(z) = zG_0(-z^{-1}) \quad (94)$$

and

$$H_k(z) = \frac{2G_k(z)}{G_k(z)G_k(z^{-1}) + G_k(-z)G_k(-z^{-1})}, \quad k = 0, 1. \quad (95)$$

The infinite impulse response (IIR) filters were implemented using the techniques in [30] following factorization of the denominator by use of the methods in [29]. Either of the above families of filters should optimize both strategies discussed in Sections III and IV, i.e., they should give optimal results when the goal is the minimization of the overall reconstruction error (strategy 1) and also when the goal is the minimization of the error component uncorrelated with the signal (strategy 2).

This is borne out by the results of a series of tests. Following [8] and [27], target rates of 0.5 b/pixel, 0.7 b/pixel, and 1.5 b/pixel were set, and the corresponding optimal bit allocations were found, for each filter used, by application of the method in [11]. The filterbanks tested for the generation of the subbands included the Haar, D2, and D6 [20], which were used in [8]. They also include two of the six state-of-the-art wavelet filter families evaluated in [27] for a target bit rate of 0.5 b/pixel. Specifically, the filters achieving best and worst PSNR in the evaluation of [27] were tested, both originally proposed



Fig. 6. (a) Reconstructed Lenna image coded with 1.5 b/pixel with the optimal analysis/synthesis filters (strategy 1). (b) Reconstructed Lenna image coded with 1.5 b/pixel with the optimal analysis/synthesis filters (strategy 2). (c) Reconstruction error for the Lenna image coded at 1.5 b/pixel using strategy 1. (d) Reconstruction error for the Lenna image coded at 1.5 b/pixel using strategy 2.

in [28]. The optimal filterbank used the filter with coefficients given in Table III for the realization of the analysis filter of the primary subband and the corresponding synthesis filter and filters of the secondary subband were found respectively by (27), (91), and (92). For the realization of the second optimal filter pair, the filter in Table III was used for the realization of $G_0(z)$ and the rest were found from (94) and (95). The bit allocation corresponding to the optimal filterbank is shown in Table IV.

The final reconstruction error variances were then determined for each filterbank. These are shown in Table V. The two optimal filter pairs produced virtually identical results.

It is seen that the optimal filters perform considerably better than the Haar and D2 filters. In comparison to the D6 filters and the filters in [28] (achieving best PSNR in [27]), the improvement in performance obtained by the use of the optimal filters is consistent at all rates and particularly pronounced at relatively low bits/pixel rates.

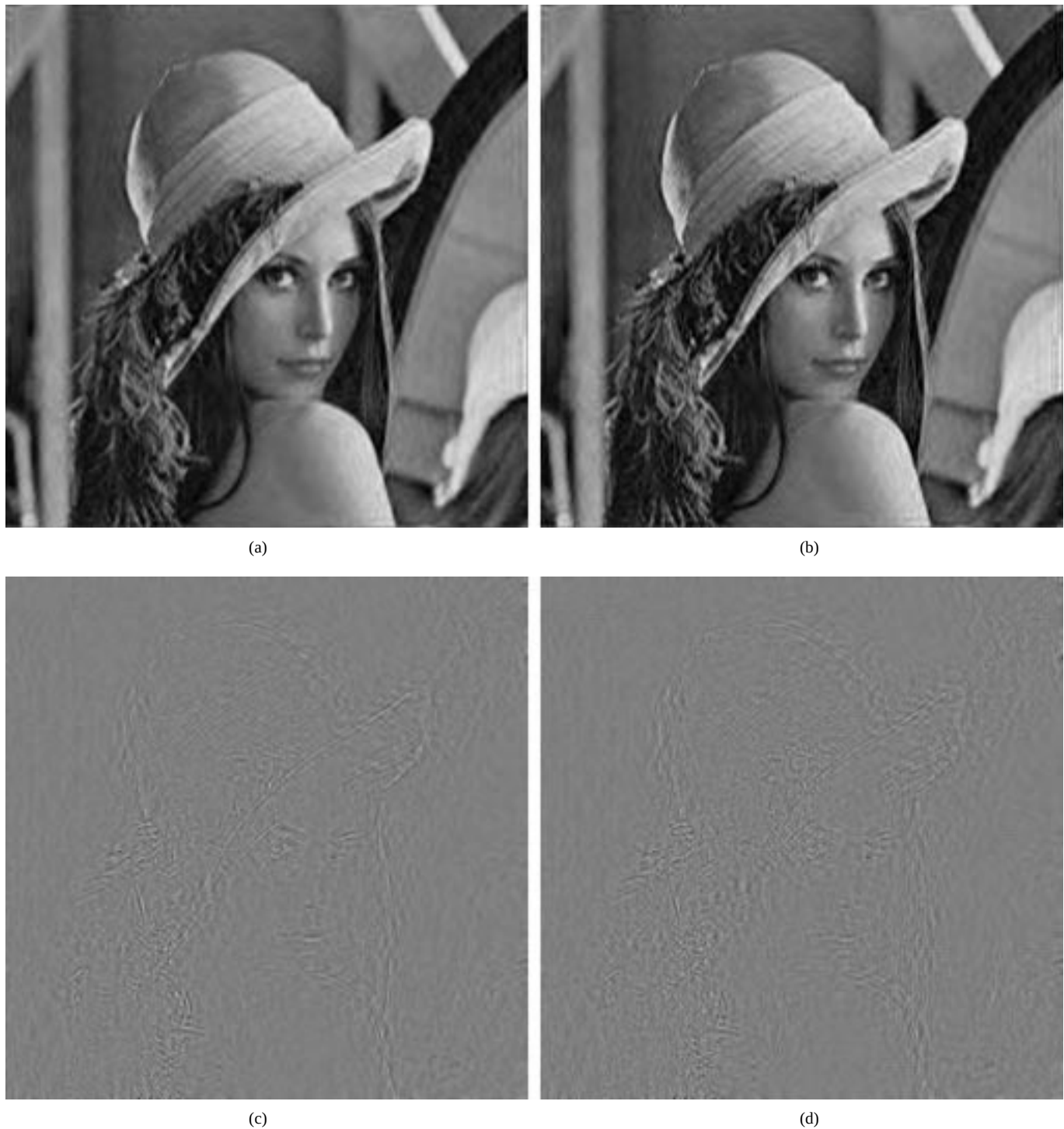


Fig. 7. (a) Reconstructed Lenna image coded with 0.7 b/pixel with the optimal analysis/synthesis filters (strategy 1). (b) Reconstructed Lenna image coded with 0.7 b/pixel with the optimal analysis/synthesis filters (strategy 2). (c) Reconstruction error for the Lenna image coded at 0.7 b/pixel using strategy 1. (d) Reconstruction error for the Lenna image coded at 0.7 b/pixel using strategy 2.

VII. CONCLUSION

A general method was developed for the analysis of the effects of quantization in subband filterbank image coders. The overall coding error, expressed as the variance of the difference between the original and the decoded image, was explicitly determined as a function of the subband filters and the quantizers used. A rigorous statistical description of the effects of a vector Lloyd–Max quantizer was also

developed. It was seen that if vector Lloyd–Max quantizers are used, the overall coding error is minimized when using the class of filters which achieve perfect signal reconstruction in lossless coding. This is generally not the case if other types of quantizers are used. However, the same methodology developed in this paper may be used for the determination of optimal filter classes corresponding to other quantizers.

For the specific case of Lloyd–Max quantizers two strategies were evaluated for the determination of the optimal coder

filters. In the first, the goal was the minimization of the overall coding error. A perfect reconstruction filterbank resulted and was optimized by choosing the analysis filters so as to minimize the overall coding error. A second strategy that has been the subject of considerable attention in recent literature was also analyzed. In this, all errors correlated with the input are set to zero, leaving a random error residue uncorrelated with the input, that may be dealt with by using random noise removal methods. The variance of the error produced by this strategy was explicitly evaluated. Assuming the use of Lloyd–Max quantizers, this strategy was optimized by selecting the analysis filters so as to minimize this error variance.

The performance of these filterbanks was evaluated experimentally, for the coding of a three-level-pyramid subband split of the 256×256 image Lenna. Assuming Gaussian statistics, the optimal vector Lloyd–Max quantizer was found and was applied to the top (lowpass) image. Optimal scalar Lloyd–Max quantization was used for the remaining subbands. It was seen that the optimal filters outperformed the Haar, D2, and D6 filterbanks, as well as the best of the filters tested in [27], when used for the realization of either of the two aforementioned coding strategies.

A number of issues remain unresolved and could be the subject of future work. First, the development in this paper was based on the simplifying restriction (20), which statistically separates the two subbands. This restriction is often desirable and in some applications necessary [31]. However, lifting this restriction may produce better results, even if the resulting filterbank turns out not to have the perfect reconstruction property. The tools developed in the present paper may help explore this possibility. Second, the removal of the random noise in strategy 2 remains an unresolved issue. Following our optimization of the synthesis filters, the remaining noise may not be further reduced by linear postfiltering, because any linear postfilter may equivalently be assumed to reside on the synthesis filters G . Since these were optimized, no further random noise reduction may be effected by linear means. This leaves unexplored, however, the option of nonlinear means for removing this noise. This option was not investigated by the present paper.

A final issue meriting further investigation regards the optimum value of the ideal bandreject analysis/synthesis filters within their passbands. This depends on the relation of the gain $A(z)$ to the variance of the quantizer input. In recent publications [9], [32], filterbanks were investigated using quantizers from the class in which the variance of the quantizer noise may be assumed to be proportional to the variance of the quantizer input. For such filter banks a constant value of the optimal filters within their passbands was found to be best, if the filter bank is preceded by a “half-whitening” [1] prefilter $C(e^{j\omega})$ satisfying $C(e^{j\omega})C(e^{-j\omega}) = 1/\sqrt{\Phi_x(e^{j\omega})}$, and followed by a reciprocal postfilter $1/C(e^{j\omega})$.

APPENDIX

The following Lemma and its Corollary were established in [10]. They are repeated here, without proof, for ease of reference.

Lemma 1: Let $x[m]$ be a zero-mean wide-sense stationary sequence with correlation sequence $R_x[p]$

$$E\{x[m+p]x[m]\} = R_x[p].$$

Let $C[k, p]$ be the covariance sequence of $x[m+p]x[m]$

$$C[k, p] = E\{x[m+p+k]x[m+k]x[m+p]x[m]\} - R_x^2[p].$$

Then either the condition

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=0}^K |C(k, p)| = 0 \quad (A1)$$

uniformly with respect to p , or the regularity condition

$$|C[k, p]| \leq C_1[k] \quad (A2)$$

where $C_1[k]$, vanishes as $k \rightarrow +\infty$, uniformly with respect to p , suffice to guarantee that $x[n]$ is correlation ergodic, as follows:

$$\lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{-K}^K x[m+p]x[m] = R_x[p] \quad (A3)$$

in the mean-square sense, uniformly with respect to p , and that if pointwise, $|\alpha[m]| \leq 1$, and

$$\lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{-K}^K \alpha[m] = 0. \quad (A4)$$

For example, if $a[m] = (-1)^m$, then

$$\lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{-K}^K \alpha[m]x[m+p]x[m] = 0 \quad (A5)$$

in the mean-square sense, uniformly with respect to p .

Corollary: If $x[m]$ is Gaussian and if $R_x[m] \rightarrow 0$ as $m \rightarrow \infty$ then (A3) and (A5) are valid.

Lemma 2: Consider a vector Lloyd–Max quantizer with zero-mean input sequence $\{v_i\}$ and output sequence $\{s_i\}$. The quantizer operates on K -dimensional vectors $[v_i, v_{i+1}, \dots, v_{i+K-1}]$ of input data. If K is sufficiently large, then i) the sequences s_i and $q_i \triangleq v_i - s_i$ are uncorrelated. If further v_i and s_i are jointly wide-sense stationary, then ii) there exists a linear time-invariant filter with impulse response a_n such that, if

$$r_m = s_m - a_m * v_m \quad (A6)$$

where the asterisk denotes convolution, then the sequences r_m and v_m are uncorrelated. The filter transfer function is given by

$$\begin{aligned} A(z) &= A(z^{-1}) = \mathcal{Z}\{a_m\} = \Phi_s(z)/\Phi_v(z) \\ &= 1 - \Phi_q(z)/\Phi_v(z) \end{aligned} \quad (A7)$$

where $\Phi_v(z)$, $\Phi_s(z)$, $\Phi_q(z)$ are respectively the power spectral densities of the input, the output and the quantization error of the quantizer, which are related by

$$\Phi_q(z) = \Phi_v(z) - \Phi_s(z). \quad (A8)$$

Finally, the power spectral density of the sequence r_m in (A6) is given by

$$\Phi_r(z) = A(z)(1 - A(z^{-1}))\Phi_v(z), \quad (A9)$$

Proof: For the coding of a K -dimensional vector $\mathbf{v} = [v_i v_{i+1} \cdots v_{i+K-1}]^T$ of input data into K -dimensional output $\mathbf{b} = [s_i s_{i+1} \cdots s_{i+K-1}]^T$, the generalized Lloyd–Max quantizer is defined by the selection of the regions B_k such that all $\mathbf{v} \in B_k$ are quantized to $\mathbf{b}_k$, $k = 1, \dots, L$ so as to minimize

$$\begin{aligned} J &= \sum_{k=1}^L \int_{B_k} \|\mathbf{v} - \mathbf{b}_k\|^2 P_v(\mathbf{v}) d\mathbf{v} \\ &= \sum_{k=1}^L \int_{B_k} [\mathbf{v} - \mathbf{b}_k]^T [\mathbf{v} - \mathbf{b}_k] P_v(\mathbf{v}) d\mathbf{v} \end{aligned}$$

where $P_v(\mathbf{v})$ is the probability density of $\mathbf{v}$ and $d\mathbf{v}$ is a short notation for the multiple differential $dv_1 \cdots dv_K$. Assuming zero mean for $\mathbf{v}$, the above is written

$$\begin{aligned} J &= \sum_{k=1}^L \int_{B_k} \mathbf{v}^T \mathbf{v} P_v(\mathbf{v}) d\mathbf{v} - 2 \sum_{k=1}^L \int_{B_k} \mathbf{b}_k^T \mathbf{v} P_v(\mathbf{v}) d\mathbf{v} \\ &\quad + \sum_{k=1}^L \mathbf{b}_k^T \mathbf{b}_k \int_{B_k} P_v(\mathbf{v}) d\mathbf{v} \end{aligned}$$

and on differentiating with respect to $\mathbf{b}$ and setting the derivative to zero, is seen to be minimum when

$$\mathbf{b}_k = \int_{B_k} \mathbf{v} P_v(\mathbf{v}) d\mathbf{v} / \int_{B_k} P_v(\mathbf{v}) d\mathbf{v} \quad (\text{A10})$$

i.e., when $\mathbf{b}_k$ is the centroid of B_k [15]. Now, if $\mathbf{q} = \mathbf{v} - \mathbf{b}$, directly from (A10)

$$\begin{aligned} E\{\mathbf{q}\mathbf{b}^T\} &= E\{(\mathbf{v} - \mathbf{b})\mathbf{b}^T\} = \sum_{k=1}^L \int_{B_k} \mathbf{v} \mathbf{b}_k^T P_v(\mathbf{v}) d\mathbf{v} \\ &\quad - \sum_{k=1}^L \int_{B_k} \mathbf{b}_k \mathbf{b}_k^T P_v(\mathbf{v}) d\mathbf{v} = \sum_{k=1}^L \left(\int_{B_k} \mathbf{v} P_v(\mathbf{v}) d\mathbf{v} \right. \\ &\quad \left. - \mathbf{b}_k \int_{B_k} P_v(\mathbf{v}) d\mathbf{v} \right) \mathbf{b}_k^T = 0 \end{aligned} \quad (\text{A11})$$

by virtue of (A10). For a more general derivation of this result, see [15]. Thus, if the dimension K of the vectors is sufficiently large it is clear that

$$E\{q_m s_k\} \cong 0 \quad \text{when } m \neq k \quad (\text{A12})$$

and part i follows. To establish part ii, note that since s_k and q_m are uncorrelated, (A6) implies that the cross correlation of r and v is given by

$$R_{rv}[m] = R_{sv}[m] - a_m * R_v[m]$$

which will become zero if

$$R_{sv}[m] = a_m * R_v[m]. \quad (\text{A13})$$

Equivalently, if the z -transform $A(z)$ of the sequence a_m satisfies

$$A(z) = \mathcal{Z}\{a_m\} = \Phi_{sv}(z)/\Phi_v(z). \quad (\text{A14})$$

Since s and $q = v - s$ are uncorrelated, then

$$\Phi_v(z) = \Phi_{s+q}(z) = \Phi_s(z) + \Phi_q(z).$$

Hence (A8). Further, $\Phi_{sv}(z) = \Phi_s(z)$. Thus, (A7) follows from (A8) and (A14). Finally, from (A6) and since v and r are uncorrelated

$$\Phi_s(z) = A(z)A(z^{-1})\Phi_v(z) + \Phi_r(z).$$

Also from (A7)

$$\Phi_s(z) = A(z)\Phi_v(z)$$

whence by equating the right-hand sides of the above, (A9) immediately results. The lemma follows. $\square$

Lemma 3: If in Fig. 3, the signal $r_0[m]$ is uncorrelated with $v_0[m]$ and $v_1[m]$, then it is uncorrelated with $x[m]$ unless

$$H_0(z)H_1(-z) - H_1(z)H_0(-z) = 0.$$

Proof: Since

$$v_k[m] = c[m]u_k[m]$$

where $c[m]$ is given by (2) and since by construction, $r_0[m] = 0$ unless m is even

$$r_0[m] = c[m]r_0[m]$$

$$\begin{aligned} E\{v_k[m+p]r_0[m]\} &= E\{c[m+p]u_k[m+p]c[m]r_0[m]\} \\ &= c[p]E\{c[m]u_k[m+p]r_0[m]\} \end{aligned}$$

where the expectations are understood in the sense of limits defined as in (3), and the obvious identity $c[m+p]c[m] = c[p]c[m]$ was used. From Lemma 1 this implies

$$E\{v_k[m+p]r_0[m]\} = \frac{1}{2}c[p]E\{u_k[m+p]r_0[m]\}$$

and if the above left-hand side is zero, this implies that for even $p = 2p_1$

$$R_{u_k r_0}[2p_1] = E\{u_k[m+2p_1]r_0[m]\} = 0.$$

Equivalently, $\Phi_{u_k r_0}(z)$ is an odd function of z . Since $\Phi_{u_k r_0}(z) = H_k(z)\Phi_{x r_0}(z)$, this results in

$$H_k(-z)\Phi_{x r_0}(-z) + H_k(z)\Phi_{x r_0}(z) = 0 \quad k = 0, 1.$$

Thus

$$(H_0(z)H_1(-z) - H_0(-z)H_1(z))\Phi_{x r_0}(z) = 0$$

whence the Lemma follows. $\square$

Lemma 4: The index

$$I = \int_{-\pi}^{\pi} (X(e^{j\omega})X(e^{-j\omega})\Phi(e^{j\omega}) - 2X(e^{j\omega})\Psi(e^{j\omega})) d\omega$$

with $\Phi(e^{-j\omega}) = \Phi(e^{j\omega}) > 0$ is minimized when

$$X(e^{-j\omega}) = \Psi(e^{j\omega})/\Phi(e^{j\omega}).$$

Proof: If $X(e^{j\omega}) + \lambda \Delta X(e^{j\omega})$ is a variation of the optimum $X(e^{j\omega})$, where λ is a real variable, the corresponding index I must be minimum at $\lambda = 0$, hence its derivative with respect to λ must be zero at $\lambda = 0$. Thus, for arbitrary $\Delta X(e^{j\omega})$

$$\int_{-\pi}^{\pi} \Delta X(e^{j\omega}) [X(e^{-j\omega})\Phi(e^{j\omega}) - \Psi(e^{j\omega})] d\omega = 0.$$

The quality within the brackets in the integrand must therefore be zero. The Lemma follows. $\square$

Lemma 5: Let $H_k(e^{j\omega})$ satisfy

$$\sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) \Phi_x(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) \times H_l(e^{-j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = 0, \quad k \neq l \quad (\text{A15})$$

$$\frac{1}{M} \sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) \Phi_x(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) \times H_k(e^{-j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = P_k \quad (\text{A16})$$

with arbitrary positive P_k . Then, the filterbank with analysis filters $H_k(e^{j\omega})$ and synthesis filters given by

$$G_k(e^{j\omega}) = \frac{H_k(e^{-j\omega}) \Phi_x(e^{j\omega})}{P_k(e^{j\omega})} \quad (\text{A17})$$

is a perfect reconstruction filterbank.

Further

$$\frac{1}{M} \sum_{k=0}^{M-1} \frac{H_k(e^{j\omega}) \Phi_x(e^{j\omega}) H_k(e^{-j\omega})}{P_k(e^{j\omega})} = 1. \quad (\text{A18})$$

Finally, if $L_k(e^{j\omega}) = H_k(e^{j\omega}) \Theta(e^{j\omega})$, where $\Phi_x(e^{j\omega}) = \Theta(e^{j\omega}) \Theta(e^{-j\omega})$, then the polyphase analysis matrix $\mathbf{F}_p$ of the filterbank with analysis filters $L_k(e^{j\omega})$ satisfies

$$\mathbf{F}_p^T(e^{-j\omega}) \mathbf{F}_p(e^{j\omega}) = \text{diag}\{P_i(e^{j\omega})\}. \quad (\text{A19})$$

Proof: The conditions for perfect reconstruction [2] are

$$\sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) G_l(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = 0, \quad k \neq l$$

and

$$\sum_{i=0}^{M-1} H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) G_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = M$$

which are obviously satisfied if (A15)–(A17) hold. Further, these imply the following matrix identity:

$$\{H_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)})\}_{k,i} \{G_l(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)})\}_{k,i} = M\mathbf{I}.$$

The above matrix multiplication with reverse order will still have $M\mathbf{I}$ as a product. This immediately implies (A18). Finally, the equivalence of (A15) and (A19) is obvious by definition of the polyphase matrix [2]. See also [22].

Lemma 6: Let $0 \leq R_k(e^{j\omega}) \leq 1$, $0 \leq \phi_k(e^{j\omega}) = \phi_k(e^{-j\omega})$, and

$$\sum_{k=0}^{M-1} R_k(e^{j\omega}) = 1 \quad (\text{A20})$$

$$\sum_{i=0}^{M-1} R_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) = 1. \quad (\text{A21})$$

Consider the functional

$$I' = \sum_{k=0}^{M-1} \int_{-\pi}^{\pi} \phi_k(e^{j\omega}) R_k(e^{j\omega}) d\omega. \quad (\text{A22})$$

Let $\Omega = \{\omega: \text{if } \mathbf{v} = \mathbf{M}^T \omega, -\pi < v_i < \pi\}$. For a given $\omega \in \Omega$, let the permutation $\{\pi_k\}$ of $\{0, 1, \dots, M-1\}$ maximize

$\sum_{k=0}^{M-1} \phi_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{\pi_k})})$. Then I' is maximized with the choice

$$R_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{\pi_k})}) = 1.$$

For example, for all ω for which

$$\sum_{k=0}^{M-1} \phi_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_k)}) \geq \sum_{k=0}^{M-1} \phi_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_{p_k})})$$

for all permutations $\{p_k\}$ of $\{0, \dots, M-1\}$, (A22) is maximum if

$$R_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_k)}) = 1, \quad k = 0, \dots, M-1.$$

Proof: As is easily seen, the index (A22) equals

$$I' = \sum_{k=0}^{M-1} \sum_{i=0}^{M-1} \int_{\Omega} \phi_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) \times R_k(e^{-j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)}) d\omega.$$

For $\omega \in \Omega$, the sets $\{\omega + 2\pi\mathbf{M}^{-T}\mathbf{q}_k\}$ and $\{\omega + 2\pi\mathbf{M}^{-T}\mathbf{q}_l\}$ are disjoint if $k \neq l$. Thus, the values of the restricted by (A20) and (A21) variables may be chosen independently for the maximization of I' . Clearly, this is accomplished when the variables R_k assume their highest values, which are equal to one, when they multiply the largest of the quantities $\phi_k(e^{j(\omega+2\pi\mathbf{M}^{-T}\mathbf{q}_i)})$. The lemma follows. $\square$

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